Field Record: SL-PADDY-MATUGAMA

Matugama, Kalutara District. Low-country Wet Zone, rainfed valley field. Cultivator K. A. Nirmala Kumari.

Document AV-FR-04 | Category: farm_record | Record current to 2 October 2026
Synthetic farm record for teaching. The fields are AgriVibe demonstration fields; cultivators, dates, observations and satellite values are invented and consistent with the AV-KB-001 thresholds.

Field details

Field id: SL-PADDY-MATUGAMA
Location: Matugama, Kalutara District
Agro-ecological zone: Low-country Wet Zone
Water regime: Rainfed valley field. Higher ground (home gardens) borders the eastern edge, where water seeps in.
Cultivated extent: 0.8 hectares
Agrarian Service Centre: Matugama
Cultivator: K. A. Nirmala Kumari
Variety: Bg 359 (3½-month age class)

2026/27 Maha season

- 5 September 2026: sowed Bg 359, broadcast method.
- 25 to 30 September 2026: heavy rainfall.
- 1 October 2026: standing water in the lower eastern section, not draining. The boundary drain along the eastern edge is partly silted.
- 2 October 2026: the cultivator reports that leaves in the lower eastern section look yellow and asks whether to add urea.

Table 1. Maha 2026/27 satellite observations

14-21 Sep 2026. Mean NDVI: 0.34; Std dev: 0.040; CV: 0.12; Uniformity: moderately_variable; Valid observations: 3; Data quality: moderate.
22 Sep-1 Oct 2026. Mean NDVI: 0.28; Change: -17.6%; Trend: significant_decline; Std dev: 0.070; CV: 0.25; Uniformity: uneven; Valid observations: 2; Data quality: moderate.

2026 Yala season (completed)

- 25 April 2026: sowed Bg 359, broadcast method.
- 8 to 20 June 2026: prolonged flooding of the field.
- 22 June 2026: the cultivator reported small reddish-brown spots on the lower leaves, starting at the leaf tips, concentrated in the eastern section next to the higher ground.
- 24 June 2026: the cultivator applied an extra urea top dressing on her own initiative. No improvement was seen by 30 June.
- 2 July 2026: the Agriculture Instructor visited, recorded symptoms consistent with iron toxicity, and advised digging a drain along the eastern boundary to intercept water seeping in from the higher ground.
- 10 July 2026: boundary drain dug. The spotting did not spread further through July.

- 15 August 2026: harvested.

Table 2. Yala 2026 satellite observations

1-8 Jun 2026. Mean NDVI: 0.58; Std dev: 0.060; CV: 0.10; Uniformity: moderately_variable; Valid observations: 3; Data quality: moderate.
15-22 Jun 2026. Mean NDVI: 0.49; Change: -15.5%; Trend: significant_decline; Std dev: 0.120; CV: 0.24; Uniformity: uneven; Valid observations: 2; Data quality: moderate.
29 Jun-6 Jul 2026. Mean NDVI: 0.47; Change: -4.1%; Trend: stable; Std dev: 0.120; CV: 0.26; Uniformity: uneven; Valid observations: 2; Data quality: moderate.